MULTI-AGENT

STOCK ANALYZER

AI-Powered News Sentiment + Technical Prediction System

Agent 1
News Sentiment

Agent 2
Technical Predictor

Agent 3
Master Synthesizer

Agent 4
Insider Intel

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THE PROBLEM

📊 Data Overload

Thousands of news articles and price data points — impossible to process manually in real time.

⏱️ Speed Gap

Stock prices react within seconds to news. Human analysis is too slow to keep up.

🧠 Cognitive Bias

Human traders fall prey to fear, greed, and recency bias — distorting judgment.

🔀 Siloed Analysis

Most tools analyze EITHER news OR technicals — never combining both into one verdict.

OUR SOLUTION

A 4-Agent AI Pipeline that reads the market like a seasoned analyst — in seconds.

AGENT 1

News Sentinel

Scrapes Yahoo, Finviz headlines
Scores sentiment -1.0 to +1.0
Detects catalysts & momentum

→

AGENT 2

Tech Predictor

RSI · MACD · Bollinger Bands
SMA 20/50/200 analysis
Price target range

→

AGENT 3

Master Mind

40% News + 60% Technical
Final BUY/SELL/HOLD verdict
Predicted price + confidence

TECHNOLOGY STACK

Groq API

Inference Engine

Ultra-fast LLM inference via Groq's LPU hardware.
Free tier with generous rate limits.

LLaMA 4 Scout

AI Model

Meta's 17B multimodal model.
Strong reasoning for financial text analysis.

yFinance

Market Data

Free historical OHLCV data.
Fundamentals, P/E, Beta, 52-week ranges.

BeautifulSoup

News Scraper

Scrapes Yahoo Finance RSS,
Finviz tables & MarketWatch feeds.

Python 3.12

Language

Runs natively in Google Colab.
No local setup required.

Google Colab

Platform

Free cloud GPU/CPU runtime.
Anywhere access, no installation needed.

SYSTEM ARCHITECTURE

DATA SOURCES

- Yahoo Finance RSS
- Finviz News Table
- MarketWatch RSS
- yFinance OHLCV
- 6-Month Price History
- Key Fundamentals

AI PROCESSING

1	**fetch_news()**	Scrape 3 free sources
2	**fetch_price_data()**	Pull 6mo OHLCV + info
3	**agent1()**	LLaMA 4 sentiment analysis
4	**agent2()**	LLaMA 4 technical forecast
5	**agent3()**	Final synthesis & verdict
6	**display_report()**	Render HTML dashboard

OUTPUT

- ✓ Final Verdict (BUY/SELL/HOLD)
- ✓ Predicted Price Target
- ✓ Confidence Score %
- ✓ Predicted Change %
- ✓ Support & Resistance
- ✓ Stop Loss Level
- ✓ Risk Assessment
- ✓ Full Reasoning

AGENT 1 — NEWS SENTIMENT ANALYST

WHAT IT DOES

- Scrapes Yahoo Finance, Finviz & MarketWatch RSS feeds
- Sends all headlines to LLaMA 4 Scout for analysis
- Scores overall market sentiment from -1.0 to +1.0
- Identifies bullish/bearish catalysts in the news
- Detects news momentum (accelerating vs. decelerating)
- Predicts bias direction for the day after tomorrow

JSON OUTPUT FIELDS

overall_sentiment	BULLISH / BEARISH / NEUTRAL
sentiment_score	-1.0 (very bearish) → +1.0 (very bullish)
confidence	0.0 – 1.0 confidence in the call
key_positives	Top bullish news drivers
key_negatives	Top bearish risk factors
catalyst_present	True/False + description
news_momentum	ACCELERATING / STABLE / DECELERATING
bias	Final direction: UP / DOWN / SIDEWAYS

AGENT 2 — TECHNICAL PRICE PREDICTOR

Relative Strength Index- RSI (14)

Identifies overbought (>70) or oversold (<30) conditions

Moving Average Convergence Divergence- MACD

Crossover signals — bullish or bearish momentum shifts

Bollinger Bands

Price volatility bands — upper/lower breakout zones

Simple Moving Average- SMA 20/50/200

Short, medium, long-term trend direction alignment

Volume Ratio

Unusual volume vs 20-day avg signals conviction

Price Momentum

1-week, 1-month, 3-month rate of change

Output: predicted_price_low · predicted_price_mid · predicted_price_high · support · resistance · stop_loss

AGENT 3 — MASTER SYNTHESIZER & FINAL VERDICT

WEIGHTING FORMULA

Combined Score = (News Sentiment Score × 0.40) + (Technical Score × 0.60)

← Score: -1.0

STRONG SELL	SELL	HOLD	BUY	STRONG BUY

+1.0 →

Final Verdict

STRONG BUY / BUY / HOLD / SELL / STRONG SELL

Price Target

Predicted price + % change from current

Price Range

Low / Mid / High target band

Confidence

Overall confidence % in the prediction

Primary Risk

Biggest risk that could invalidate call

Opportunity

Key upside driver to watch

AGENT 4 — INSIDER & INSTITUTIONAL INTELLIGENCE

WHAT IT DOES

- Fetches CEO/CFO/Director buy & sell transactions via yFinance
- Pulls top hedge fund & institutional holder positions
- Scrapes OpenInsider.com for cluster buy signals
- Reads SEC EDGAR Form 4 filings (official disclosures)
- Calculates buy/sell ratio & net insider money flow
- Detects CLUSTER BUY: 3+ insiders buying = 🚨 strong signal
- Weights 30% in final verdict alongside news & technicals

FREE DATA SOURCES

yFinance Insider	Insider transactions (buy/sell/exercise)
yFinance Institutional	Top hedge fund & mutual fund positions
OpenInsider.com	Cluster buy/sell scraping & history
SEC EDGAR Form 4	Official mandatory insider disclosures

JSON OUTPUT FIELDS

insider_signal	STRONG BULLISH → STRONG BEARISH
insider_score	-1.0 (heavy selling) → +1.0 (heavy buying)
cluster_buy_detected	True/False — triggers +0.15 score boost
smart_money_flow	ACCUMULATING / DISTRIBUTING / NEUTRAL
buy_sell_ratio	Dollar value of buys ÷ sells
red_flags	Suspicious selling patterns flagged

WHY INSIDER SIGNALS MATTER — THE SMART MONEY EDGE

Cluster Buy = #1 Signal

When 3+ insiders buy simultaneously it historically predicts 15-25% outperformance over 12 months. No other signal matches this reliability.

CEO/CFO Buying Own Stock

Executives who buy open-market shares with their own money signal deep conviction. They only buy when they believe the price is undervalued.

Form 4 = Legal Obligation

SEC Form 4 must be filed within 2 business days of any insider trade. This is official, verified data — not rumor or speculation.

Institutional Accumulation

When Vanguard, BlackRock, or top hedge funds increase positions, it signals long-term confidence backed by deep due diligence teams.

Routine Selling ≠ Bearish

Most insider selling follows 10b5-1 pre-planned schedules for diversification. Agent 4 distinguishes routine selling from panic selling.

Smart Money Leads Price

Insider buying historically leads share price by 2-4 weeks. Agent 4 catches this signal BEFORE the market reacts — giving early edge.

UPGRADED: 4-AGENT FUSION ARCHITECTURE

OLD (3-Agent Formula)

40% Technical + 60% News = Final Score

NEW (4-Agent Formula)

40% Technical + 30% News + 30% Insider = Final Score

AGENT 1

News Sentinel

Yahoo · Finviz · MarketWatch
Sentiment -1.0 → +1.0
30% weight

AGENT 2

Tech Predictor

RSI · MACD · Bollinger
SMA 20/50/200
40% weight

AGENT 4

Insider Intel

Insiders · Hedge Funds
SEC Form 4 · OpenInsider
30% weight (+0.15 cluster boost)

AGENT 3

Master Mind

Synthesizes all 3 signals
Final BUY/SELL/HOLD
Predicted price + confidence

CLUSTER BUY BOOST: If 3+ insiders buy simultaneously → combined score gets +0.15 bonus — the strongest historical signal

⚖️ ACCURACY, RISKS & LIMITATIONS

✅ STRENGTHS

- Combines two independent data streams (news + price history) for cross-validation
- Removes emotional bias — purely data-driven
- Processes months of price data instantly
- Provides explicit confidence scores
- Flags when agents disagree (conflict signal)
- Free to run — no paid data feeds needed

⚠️ LIMITATIONS

- Cannot predict black swan events (earnings surprises, geopolitical shocks)
- News scraping may miss paywalled content
- LLM hallucination risk on price targets
- Not a substitute for professional advice
- Past technical patterns don't guarantee future results
- Groq free tier has rate limits (may need delays)

FUTURE ENHANCEMENTS

Phase 1
Near-Term

- Add SEC filing reader (10-K / 10-Q Agent)
- Real-time WebSocket price streaming
- Email / Telegram alert when BUY signal fires
- Portfolio watchlist — scan 10 tickers at once

Phase 2
Mid-Term

- Options chain analysis agent
- Earnings call transcript sentiment
- Insider trading pattern detector
- Backtesting module — verify past predictions

Phase 3
Long-Term

- Agent memory across sessions
- Fine-tuned model on financial corpus
- Web dashboard with interactive charts
- Multi-stock portfolio optimizer

USE CASES & TARGET USERS

Learning Traders

Students & beginners who want to understand how AI can combine fundamental news analysis with chart technicals.

Researchers

Finance & AI researchers studying LLM performance on market prediction tasks.

Retail Investors

Self-directed investors who want a quick daily pre-market briefing on their watchlist.

Dev Builders

Python developers who want a starting framework to build production stock screening tools.

Finance Bloggers

Content creators who need quick data-backed commentary on hot stocks.

Educators

Teachers using it as a live classroom demo of multi-agent AI systems.

HOW TO RUN IN GOOGLE COLAB

01

Get Free Groq API Key

Visit console.groq.com → Sign Up → API Keys → Create Key
Copy your gsk_... key (free, no credit card)

02

Open Google Colab

Go to colab.research.google.com
Create a new notebook

03

Install Dependencies

```
!pip install groq yfinance requests beautifulsoup4 pandas numpy -q
```

04

Paste Agent Code

Copy the full multi-agent Python script
into Cell 2 and run it

05

Set API Key

Replace 'gsk_your_key_here' with your actual Groq key

06

Analyze Any Stock

Run: analyze_stock('NVDA')
Supports: AAPL, TSLA, MSFT, GOOGL, META, AMZN...

Total setup time: Under 3 minutes · Cost: $0 (free Groq tier)

START ANALYZING

SMARTER TODAY.

Three agents. Two data streams. One clear verdict.

Get Groq API Key

console.groq.com

Open Google Colab

colab.research.google.com

Try It Now

analyze_stock('AAPL')

⚠️ This tool is for educational purposes only. Not financial advice. Always do your own research.